

Subsidizing the tragedy of the commons: Fuel subsidies modify fishing behavior and drive overfishing

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Abstract

Fuel subsidies to the fishing industry are widely considered to be one of the leading drivers of overfishing, which has prompted calls for a global reform. However, rigorous empirical evidence quantifying the behavioral and production responses to fuel subsidies remains limited. We address this gap by analyzing 14 years of high-resolution data on fisher-level fuel subsidy allocations, fishing activity, and production in Mexico's shrimp trawl fleet. Leveraging year-to-year variation in Mexico's subsidy policy, we find that receiving a fuel subsidy increases fishing effort by 40.7%, fished area by 23.5%, and landed catch by 70.2%. These responses scale with the size of the subsidy: a 1% increase in the subsidy received raises fishing effort, area fished, and landings by 0.14% to 0.20%. Mexico's abrupt subsidy elimination in 2020 reduced fishing pressure through two channels: a 30% reduction in fleet size as marginal economic units exited, and lower fishing time among the economic units that remained. Together, these results identify fuel subsidies as a major driver of fishing pressure: roughly 27% of

26 fishing effort and 40% of landings in this fleet are attributable to subsidies, supporting
27 ongoing global efforts to eliminate them.

28 1 Introduction

29 Fisheries subsidies are widely regarded as a major driver of overfishing [1, 2]. This is partic-
30 ularly true for fuel subsidies, because subsidizing an input lowers its effective price and leads
31 to inefficiently high input use relative to the social optimum. In the context of fisheries,
32 fuel subsidies result in two sources of lost economic efficiency [3]. The first is the usual cost
33 associated with a market distortion, which arises because resources are being misallocated.
34 The second is associated with greater production of the externality, in this case overfishing.
35 This paper focuses on this second type of deadweight loss in the context of fuel subsidies in
36 industrial fisheries.

37 Fisheries subsidies are prevalent in most coastal nations, where governments typically
38 spend enormous amounts on subsidy programs [1, 3]. In 2018 alone, global fisheries subsidies
39 totaled approximately USD \$35.4 billion, of which USD \$7.7 billion were allocated specifically
40 to fuel subsidies [4]. These large figures have prompted calls for global subsidy reforms [5, 6],
41 and a particular focus has been placed on ending cost-reducing and capacity-enhancing
42 subsidies such as fuel subsidies. Although there is broad consensus on the potential benefits
43 of subsidy reforms [2], empirical evidence on the magnitude of these benefits remains limited.

44 Our ability to predict the social and environmental consequences of fuel subsidy reforms
45 depends on answering two central unresolved questions: How much *additional* fishing effort
46 do fuel subsidies induce? and *where* does that effort materialize? If the amount of overfishing
47 induced by fuel subsidies is large, then the reforms could have large upsides. However, if the
48 amount is small relative to other sources of overfishing (*e.g.* by-catch or illegal, unreported,
49 and unregulated fishing), then it may be better to focus management efforts on addressing
50 those. Similarly, if most “subsidized” effort is spatially concentrated in a few regions, then we
51 would expect that only those regions would benefit from a reform [7]. Here, we address these

two questions by analyzing 14 years of high-resolution vessel tracking data from Mexican shrimp trawlers and long-term administrative records on fisher-level subsidy allocations.

There are two empirical challenges to studying the effect of fuel subsidies on fishing outcomes. The first is data availability. One must observe subsidy allocations, fishing behavior, and/or fisheries production through time, all of which are rarely available at the vessel-level. And even when they are, a second challenge arises: Governments typically implement blanket subsidy policies for all fishers in a region, which eliminates the quasi-experimental variation required to establish causality in observational studies [8].

However, recent improvements in data and research design have begun to yield credible evidence on this question. For example, Wang et al. [9] studied the extensive-margin effects of subsidies by studying the interaction between fuel subsidies and vessel buyback programs in China. They found that reductions in fuel subsidies accelerated the rate at which fishermen retired their vessels, but were not able to study intensive-margin effects. Recent work by Englander et al. [8] leveraged vessel-tracking data to show that fuel subsidies to China’s distant water fishing fleet have a large impact on the fleet’s fishing effort. While they were able to study the intensive and extensive margin effects of fuel subsidies, they did not directly observe vessel-level subsidy allocations.

To the best of our knowledge, these studies are the only ones to provide rigorous causal estimates of the effect of fuel subsidies on fisheries. Important as their contributions are, they focus on China, a nation whose fishing dynamics, magnitude, and scale are so different that most work on fisheries treats them differently [10, 11]. Furthermore, neither work studied the effects on fisheries production or changes in the spatial footprint of fishing, further limiting our ability to understand how much biomass is being extracted from the system, and which areas may benefit the most from a reform.

We make three contributions. First, our work is among the first ones to directly observe vessel-level subsidy allocations alongside fishing behavior and production, rather than inferring exposure from policy variation. Second, we move beyond effort to quantify how

subsidies shape both fisheries production and the spatial footprint of fishing, revealing not just how much extraction subsidies induce, but where. Third, we exploit Mexico’s abrupt and complete 2020 subsidy elimination as a natural experiment, allowing us to estimate responses to a full reform of the kind the WTO agreement now contemplates.

Mexico is the world’s 13th largest fishing nation [11] and provides an ideal empirical setting due to its long history of subsidizing fisheries, particularly through fuel subsidies [12, 13, 14, 15]. We focus on Mexico’s shrimp trawl fleet because they are Mexico’s largest and most important fleet, accounting for more than 45% of the value of all Mexican fisheries and catching around 50,000 tons each year [16, 17, 18]. Fuel represents around 40% of the fleet’s variable costs, and subsidies are an important source of their profit [19, 15]. The fleet also received between 48% and 67% of the annual fuel subsidies provided to all industrial economic units fishing in Mexico.

The paper proceeds as follows. Section 2 provides key details on Mexico’s fuel subsidy program, explains our identification strategy, and provides an overview of our panel data set. Section 3 presents our results, which are discussed in Section 4. A detailed account of our data and methods are provided in Section 5.

2 Mexico’s Fuel Subsidy Program

Between 2007 and 2019, Mexico’s fishery management agency (CONAPESCA) maintained a limited-entry roster of economic units eligible to receive a fuel subsidy each year. An economic unit may only “enter” the roster if another unit “exits” it¹. Once in the roster, the amount of subsidy granted to an economic unit is a function of their vessel’s engine power and the prescribed duration of the fishing season. However, fishery managers can adjust the final subsidy amount each unit receives based on an “adjustment factor”, a constant across all units that depends on the annual national allocations to the program (For more details,

¹For example, an economic unit could be removed from the roster if they failed to carry a functioning vessel monitoring system or if they failed to report their landings.

see Appendix A.1). Subsidized economic units receive money via a government-issued debit card, which can only be used at fueling depots. Any unspent money at the end of the year is reclaimed by CONAPESCA.

It is worth mentioning that the fuel subsidy program was suddenly eliminated in 2020 after Mexico’s former president signed into law the “Republican Austerity Act” as a way of curtailing public spending in the face of the COVID-19 pandemic [15]. Although this led to multiple protests from fishers and increased political pressure², the federal government maintained its position and Mexican fishers have not received subsidized fuel since.

2.1 Identification Strategies

We leverage two complementary sources of quasi-experimental variation introduced by Mexico’s fuel subsidy program design, as well as the *impromptu* subsidy reform. Together, these provide plausibly exogenous shocks to subsidy allocations that allow us to estimate how fuel subsidies shape fishing behavior and fisheries production. We summarize the core identifying assumptions underlying each source of variation and the corresponding estimation strategy.

Changes in subsidy status. First, we exploit changes in subsidy status generated by the institutional design of CONAPESCA’s subsidy roster. Entry into the roster can only occur when another economic unit exits. Because an economic unit cannot directly influence whether others exit the roster, these transitions effectively induce externally driven shifts between *subsidized* and *not subsidized* status for units that are subsidized only in some years. We use this roster turnover to identify how receiving a subsidy affects the probability of fishing, the probability of reporting landings, and the magnitude of fishing activity conditional on participation in the fishery. These effects are estimated using two-way fixed effects regressions that capture semi-elasticities with respect to subsidy status.

²See El Sudcaliforniano: Pega a pescadores la falta de apoyos for a news report and a letter by Senator Cecilia Sánchez García denouncing the removal of fisheries subsidies.

Changes in subsidy amounts. Second, we leverage within-unit variation in subsidy amounts generated by the program’s annual “adjustment factor” revisions. This factor is determined by federal budget allocations and administrative decisions that vary across regions and years but are orthogonal to individual fishers’ short-run incentives or expected fishing behavior. As a result, the year-to-year fluctuations in subsidy amounts for the same economic unit provide treatment-intensity variation that is plausibly exogenous. For example, Figure S2 shows how economic-unit-level subsidy allocations changed between 2011 and 2019, when Mexico provided fuel subsidies to its fleet. We exploit this within-unit, across-year variation using log–log panel data regressions with economic-unit and region-year fixed effects, yielding elasticities of fishing time, fished area, and landed catch with respect to the amount of subsidy received.

Abrupt elimination of subsidies in 2020 Third, we exploit the sudden and unanticipated elimination of fuel subsidies in 2020 as a natural experiment. Because the reform was nationwide, non-discretionary, and unrelated to the performance or behavior of individual economic units, it provides a clean policy shock. We use this setting to estimate (i) the probability that an economic unit exits the fishery after subsidies disappear and (ii) changes in fishing behavior among units that remained active. These effects are estimated with two-way fixed effects regressions.

Together, these three sources of variation allow us to estimate: (i) The percent change in fishing activity with respect to changes in subsidy status; (ii) The percent change in fishing activity associated with a 1% increase in the subsidy amount; and (iii) long-run and aggregate responses following the 2020 reform. To do so, we combine different sources of administrative data to construct a novel annual panel data set on subsidy allocations (MXN \$), time fishing (hours), area fished (km²), and landed catch (kg) by economic unit (i.e. an individual fisher or fishing company). This panel allows us to observe the annual fishing behavior, spatial footprint, and fisheries production of 920 shrimp trawling vessels owned by

341 economic units. See Section 5.1 for details on data source and sample construction.

As an example to further motivate our analysis, Figure 1 shows how fishing activity by one economic unit changes when they receive a fuel subsidy of MXN \$231,543 (about USD \$12,388). When subsidized, the economic unit increases fishing effort (from 117 days to 303 days) and the extent of its fishing grounds (from 24,168 km² to 33,338 km²). Of course, this example from a single economic unit cannot control for confounding factors, but it shows how fuel subsidies might lead to an intensification and an expansion of fishing effort.

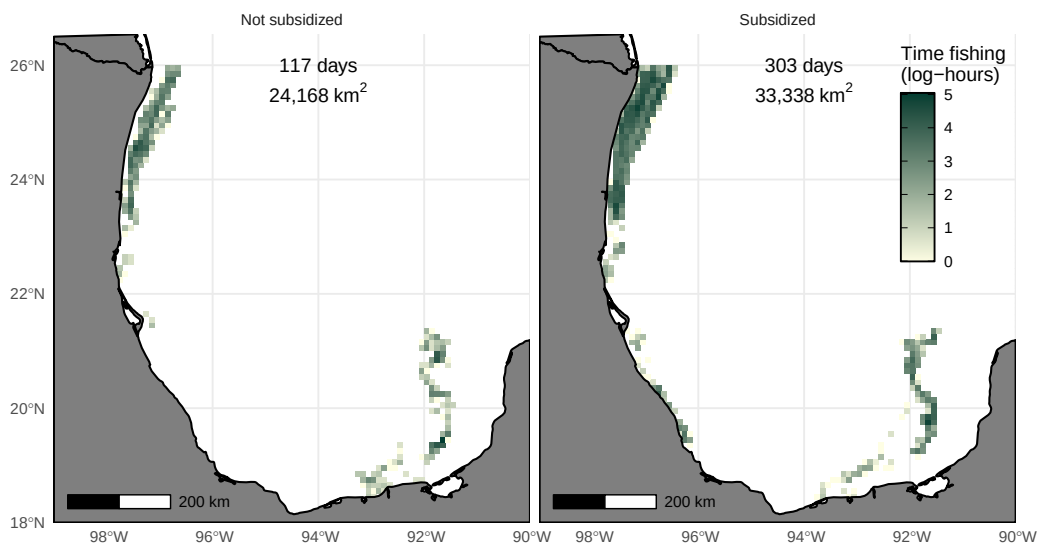


Figure 1: Example of changes fishing behavior in relation to subsidy status. Maps show fishing activity by the same economic unit in a year without a subsidy (left) and a year with a subsidy (right). This fisher spent nearly three times more time fishing when subsidized than when not, and the extent of its fishing grounds is around 50% larger when subsidized than when not. The footprint of fishing effort is shown along a 0.1° grid.

3 Results

We begin with a visual inspection of our three outcomes of interest (time fishing, area fished, and landings) in Figure 2. This exercise reveals three general patterns. First, economic units spend more time fishing, fish a greater area, and land more shrimp when they are subsidized. Second, economic units that were always subsidized between 2011 and 2019 ($N = 142$) consistently fish more than those that were only sometimes subsidized ($N = 167$).

And third, that the pattern persists even for the subset of economic units whose subsidy status changes through time (those labeled “sometimes”).

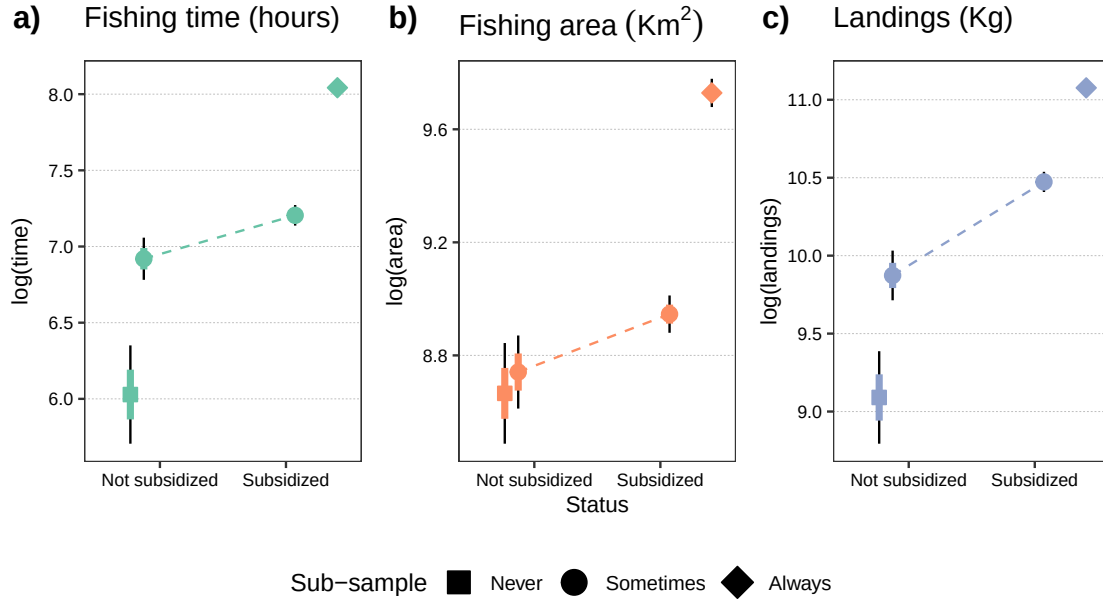


Figure 2: Fishing behavior and fisheries production in relation to subsidy status (2011-2019). The horizontal axis shows the subsidy status and the vertical axis shows the log-transformed outcome of interest. Points show mean values, error bars show standard errors (colored portion) and 95% confidence intervals (thin black lines). Marker shapes indicate subsidy category with respect to number of times subsidized (never, sometimes, always). The dashed lines connect the mean values for economic units that are sometimes subsidized across subsidized status.

We structure our results around the three sources of variation described in subsection 2.1. First, we examine how changes in subsidy status affect fishing behavior and production. We then quantify how fishing responds to the *amount* of subsidy received. Finally, we study changes in fishing behavior following the 2020 subsidy reform. At the end, we use our estimates to perform a retrospective analysis that quantifies the historical and spatial influence of Mexico’s fuel subsidy program on the shrimp trawl fleet.

3.1 Responses to change in subsidy status

Our sample contains 341 economic units, of which 32 never received a subsidy, 142 were always subsidized, and 167 were subsidized only in some years. We exploit within-unit changes in subsidy status for this last group of economic units to estimate how receiving

a subsidy affects participation, fishing activity, and landings. Using a two-way fixed-effects regression (see Methods in 5.3.1), we find that receiving a subsidy increases the probability that an economic unit engages in fishing by approximately 22 percentage points, as measured by both fishing activity and area fished. The probability of reporting landings increases by 41 percentage points ($p < 0.01$; Table 1A).

We then examine whether an economic unit fishes more when they are subsidized. On average, subsidized units fish 350 additional hours per year ($p < 0.01$), expand their fishing grounds by 2,528 km² ($p < 0.01$), and land 32.3 additional tons of shrimp ($p < 0.01$; Table 1B). Relative to the mean outcomes of unsubsidized units, these correspond to increases of 24.8%, 36.5%, and 170% in time fishing, fishing area, and landings, respectively.

Finally, we estimate semi-elasticities for economic units that always have positive fishing effort, regardless of their subsidy status ($N = 134$, a subset of the sometimes-subsidized units). For these, receiving a subsidy is associated with a 40.7% increase in time fishing ($p < 0.01$), a 23.5% expansion in fishing grounds ($p < 0.01$), and a 70.2% increase in landings ($p < 0.01$; Table 1C). Because economic units can come in and out of the roster multiple times, we also use a difference-in-differences estimator robust to treatment switching following de Chaisemartin and D’Haultfœuille [20]. Our results are robust to alternative specifications, sample definitions, and estimators (see Table S2–Table S4 and Figure S5–Figure S7).

Table 1: Effect of receiving a fuel subsidy on fishing behavior and fisheries production.

	Fishing time	Fishing area	Landings
A) Extensive margin			
Received subsidy	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Intensive margin (levels)			
Received subsidy	350.269 (65.646)***	2528.340 (522.980)***	32347.423 (6977.661)***
$\bar{Y}_{\text{Subsidized}=0}$	1411	6938	18968
N_{eu}	167	167	167
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
C) Semi-elasticities (logs)			
Received subsidy	0.342 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
%Change	40.71%	23.49%	70.18%
N_{eu}	134	127	117
N	1290	1235	1192
R^2 Adj	0.726	0.699	0.757

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows estimates for the extensive margin, where the outcome variables indicate whether a vessel spent time fishing, had fishing grounds, or reported landings. Panel B) shows estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km²), and landings (kg). Panel C) shows semi-elasticity estimates for log-transformed time fishing (hours), fishing area (km²), and landings (kg). This last panel excludes observations with fishing activity or landings at exactly zero, mostly capturing the intensive margin.

3.2 Responses to change in subsidy amount

A key feature of Mexico’s fuel subsidy program is that the annual subsidy amount allocated to each economic unit varies across years (see Figure S2). Because these adjustments are driven by budgetary conditions rather than unit-specific factors, they generate plausibly exogenous within-economic-unit variation in subsidy amounts through time. We use this variation to estimate intensive-margin responses for 297 economic units that were subsidized at least twice between 2011 and 2019. Using two-way fixed-effects log-log regressions, we estimate elasticities of fishing time, fished area, and landings with respect to subsidy amount. We find that a 1% increase in the subsidy received is associated with a 0.14% increase in time fishing ($p < 0.01$), a 0.15% increase in fished area ($p < 0.01$), and a 0.20% increase in landings ($p < 0.01$). These estimates are robust to alternative sample definitions and model specification (see Table S5 and Figure S10).

Table 2: Elasticity estimates for time fishing (hours), fishing area (km²), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
%Change	0.14%	0.15%	0.20%
N_{eu}	297	292	295
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. All models include fixed effects by economic unit and by region-year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. The sample contains economic units subsidized at least twice and with subsidy amount > 0 . The number of economic units used in each column is shown by N_{eu} .

3.3 Responses to an *impromptu* reform

Our results thus far rely on within-program variation to estimate short-run changes in fishing behavior. Recall that Mexico abruptly eliminated its fuel subsidy program [15], which prompted protests from fishers who argued they could not continue operating without financial support [21]. This intervention provides an opportunity to study how economic units respond when subsidies disappeared entirely.

We begin by examining the exit behavior across the fleet. We focus on the 142 economic units that were always subsidized prior to the reform. All were consistently active between 2011 and 2019, yet 53 failed to participate in the fishery at least once between 2020 and 2024, and 42 did not return at all by 2024, suggesting permanent exit of roughly 30% of the always-subsidized fleet (Figure 3). Estimating the probability of exit after 2020, we find that the average annual probability of an economic unit leaving the fishery increased by 7.5 percentage points in the post-reform period ($p < 0.01$; Table 3). Among units that continued fishing after the reform, we estimate that average annual fishing time fell by 4,200 hours, fished area declined by 12,900 km², and landings decreased by 45.6 tons (Table 4). Together, these results indicate that the reform reduced total fishing effort through two channels: the permanent exit of marginal economic units, and the reduction in effort among those that continued to fish.

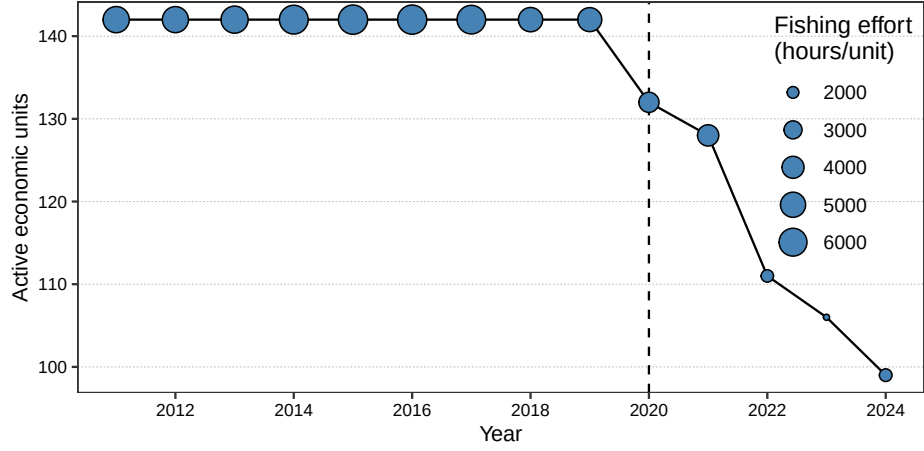


Figure 3: Some economic units exit the fishery after the subsidy reform, and those who stay fish less. The horizontal axis shows time in years. The vertical axis shows the number of economic units participating in the fishery. The vertical dashed line marks 2020, the year in which the subsidy program was removed. The size of points indicates the mean fishing effort (hours per economic unit) for a given year. The figure contains data for economic units that were always subsidized between 2011 and 2019 ($N = 142$).

Table 3: Effect of Mexico's *im-promptu* fuel subsidy reform on probability of economic units exiting the fishery.

(1)	
After reform	0.075 (0.011)***
N	1914
R^2 Adj	0.049

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level.

Table 4: Effect of Mexico’s *impromptu* fuel subsidy reform on fishing behavior and fisheries production for economic units that did not exit the fishery.

	Fishing time	Fishing area	Landings
After reform	−4273.62*** (685.28)	−12 929.86*** (1067.87)	−45 651.01*** (10 855.58)
$\bar{Y}_{\text{Post}=0}$	4472	18611	107863
N	1400	1400	1400
R^2 Adj	0.783	0.762	0.868

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Table shows coefficient estimates for the intensive margin, where the outcome variables are time fishing (hours), fishing area (km^2), and landings (kg).

3.4 Aggregate effects of fuel subsidies

We have shown that subsidized economic units fish more, and that the amount of additional fishing increases with the amount of subsidy received. We now examine what these *individual* responses imply for *aggregate*, fishery-wide impacts. In particular, we ask: How much of total historical effort is attributable to Mexico’s fuel subsidies program? And, was this effort disproportionately applied in specific areas?

3.4.1 Historical impacts of subsidies

Total annual fishing activity (or landings) can be divided into three categories: (1) activity by units that were never subsidized; (2) activity by subsidized units that would have occurred even without the subsidy; and (3) activity by subsidized units that is attributable to the subsidy. We identify this last category by leveraging our semi-elasticity estimates to retrospectively simulate a counterfactual with no fuel subsidies disbursed between 2011 and 2019. Because our estimating equation is $\log(y_{it}) = \beta S_{it} + \omega_i + \lambda_{r(i)t} + \epsilon_{it}$, the activity we observe for a subsidized unit is $\exp(\beta)$ times what that same unit would have done had it not been subsidized. The counterfactual is therefore recovered by scaling observed activity by $\exp(-\beta)$, so the share of *observed* subsidized activity that is attributable to the subsidy is $1 - \exp(-\beta)$. For fishing effort ($\beta = 0.342$) that share is 28.9%. Note that this is the same estimate expressed against a different base: receiving a subsidy raises effort by 40.7% *relative to the counterfactual* (Table 1C), which is equivalent to 28.9% *of the observed, subsidized level*. For every year in our data, we identify the economic units that were subsidized in that year and scale their fishing effort down accordingly. Importantly, because the number and identity of economic units subsidized changes through time, this is not the same as simply reducing total effort by 28.9%.

Between 2011 and 2019, Mexican shrimp trawlers fished an average of 1.14 ± 0.15 million hours, of which 1.05 ± 0.15 million were by subsidized units (Figure 4a). Using our semi-elasticity estimates, we find that 0.30 ± 0.04 million hours per year can be attributed to

fuel subsidies, implying that subsidies accounted for $26.7\% \pm 1.7\%$ of total annual fishing hours. The fleet landed an average of 25.0 ± 2.3 thousand tons of shrimp (live weight) per year, with 24.0 ± 2.6 thousand tons landed by subsidized units (Figure 4c). Our calculations indicate that 9.9 ± 1.1 thousand tons of landings—about 40% of all shrimp landed by the studied fleet—were attributable to subsidies.

We repeat this exercise using our elasticity estimates to calculate the *percent* reduction in fishing time and landings under alternative subsidy-reduction scenarios (10, 30, 50, and 90%). For instance, capping subsidy allocations at 50% of historical levels would have reduced fishing time by approximately 96.3 thousand hours per year and landings by 3.08 thousand tons (Figure 4b,d).

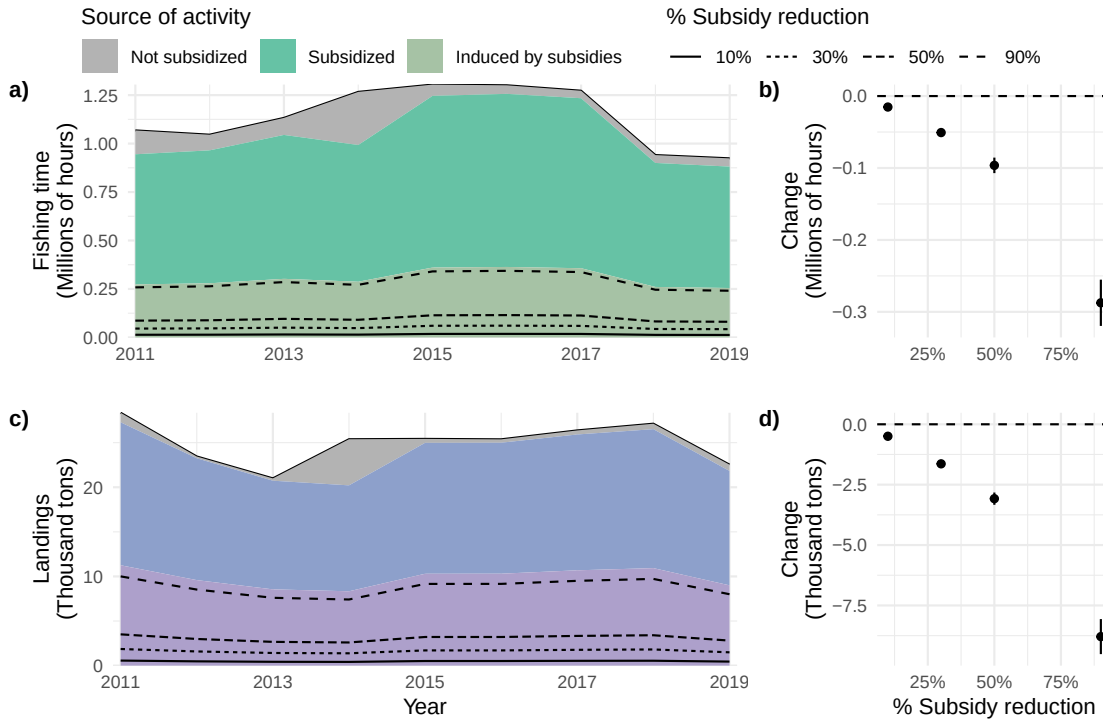


Figure 4: Aggregate effects of fuel subsidies on fishing activity, fished area, and landings. Panels a) and c) show area-stacked time-series of fishing time and landings³. The gray portion is activity and production from economic units that were not subsidized in a given year. The colored portion corresponds to activity and production by economic units that were subsidized. The bottom stack of each panel shows the portion of effort or production by subsidized economic units that is attributable to the subsidy, as indicated by our semi-elasticity estimates (Table 1C). The different line types show the portion of effort that could have been removed had the subsidies been reduced by different amounts. Panels b) and d) show the mean annual reduction in fishing time and landings expected from four different subsidy reduction policies (estimated as mean of all activity between 2011-2019). Black error bars show 95% confidence intervals and the colored portion shows standard errors.

3.4.2 Spatial implications of a non-spatial policy

The above thought experiments cannot be conducted for fished area because the metric is not additive across economic units. However, understanding the spatial implications of a non-spatial policy is crucial, as the distribution of fishing effort can strongly influence environmental impacts and determine where policies might yield upsides [22, 7]. For example, if most economic units fish in a small subset of grounds, eliminating fuel subsidies would generate large, localized upsides. Conversely, if subsidized and unsubsidized vessels operate in broadly similar areas, subsidy reform would yield modest but spatially widespread effects. This tension between large and local *versus* modest and widespread benefits raises a natural question: is subsidy-induced fishing effort homogeneously distributed in space?

To answer this, we use our semi-elasticity estimates to simulate counterfactual fishing activity in the absence of subsidies along a $0.1^\circ \times 0.1^\circ$ grid (roughly 11 km by 11 km at the equator). Pixels fished only by unsubsidized units show no change, while those fished exclusively by subsidized units exhibit the largest reductions. Using data from the final year of subsidies (2019, with 286 active units, 247 of which were subsidized), we find that subsidized fishing activity is heterogeneously distributed in space, but that this pattern broadly matches the baseline distribution of activity by unsubsidized units (Figure 5a–b).

Mexico divides its waters into six broad management regions. The Gulf of California (region II) and Gulf of Mexico (regions V and VI) sustain the highest levels of both total and subsidized activity (Figure 5a–b). Subsidized effort accounted for up to 29% of total activity along the eastern margin of the Gulf of Mexico and the Campeche Bank, between regions V and VI (Figure 5c). These areas also consistently ranked among the top 90% of pixels where subsidized effort represented a large share of total fishing (Figure 5d).

This analysis also reveals that several areas of particular conservation concern were fished primarily by unsubsidized units. For example, all activity within and around the recently protected Alacranes Reef [23] (region VI) was attributable to unsubsidized economic units. Similarly, fishing in the upper Gulf of California—region II, habitat of the critically en-

288 endangered Vaquita [24, 25]—was also predominantly conducted by unsubsidized units. This
 289 shows that eliminating subsidies would not have reduced fishing pressure in two of Mexico’s
 290 most critical habitats.

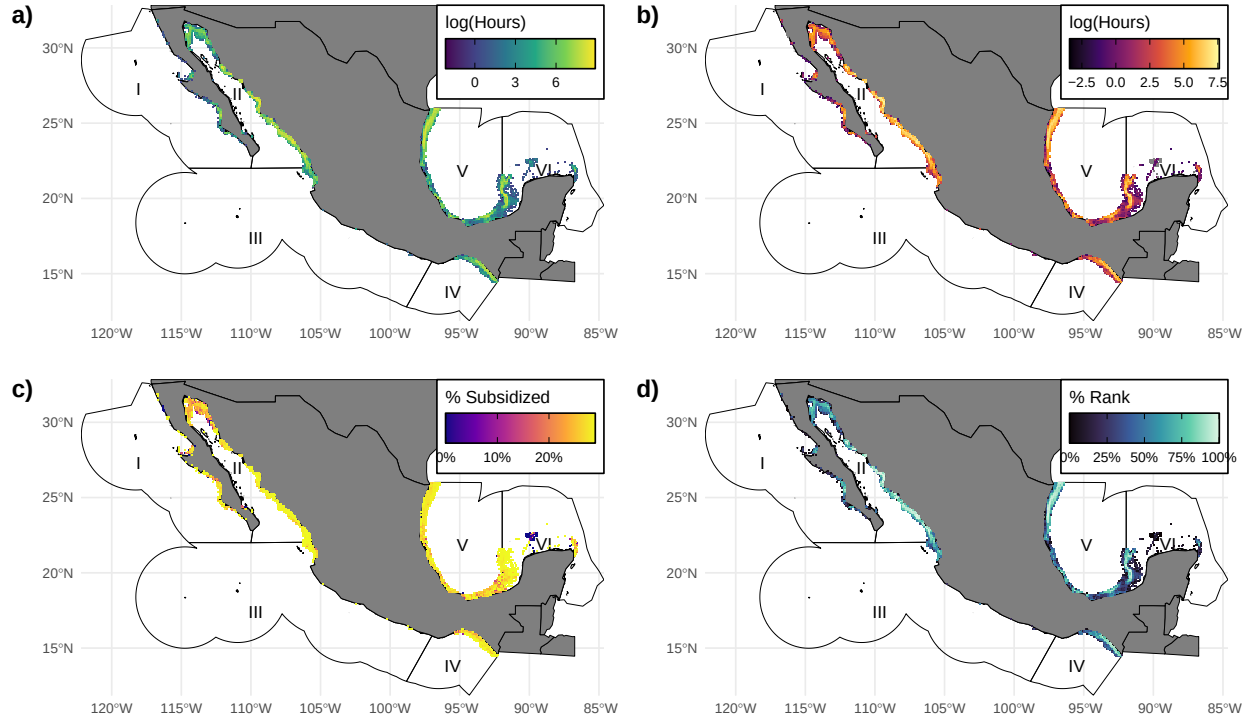


Figure 5: Spatial distribution of the effects of fuel subsidies on fishing activity. Panel a) shows a map of total fishing hours for 2019. Panel b) shows a map of total fishing hours attributable to fuel subsidies. Panel c) shows the per cent of fishing effort that is attributable to fuel subsidies, and panel d) shows the percentile ranking of each pixel. Polygons in the ocean show Mexico’s Exclusive Economic Zones, divided into six management regions utilized by Mexico’s fishery management agency.

4 Discussion

Our results show that fuel subsidies substantially increase fishing effort, the spatial footprint of fisheries, and landed catch in Mexico’s industrial shrimp fleet. More importantly, we show that the *amount* of subsidy also matters: fishing effort, fishing grounds, and landings all scale positively with the size of the subsidy allocation. We also show that Mexico’s abrupt 2020 reform led to substantial exit from the fishery and to marked reductions in effort and landings among economic units that remained active. Taken together, these findings indicate that fuel subsidies have meaningful and measurable fishery-wide impacts.

Our study has limitations common to observational research. The estimates reported in this study can be interpreted as causal effects under two conditions: first, that annual subsidy variation reflects federal budget processes rather than unobserved fisher characteristics, and second, that our outcomes of interest are largely immune to manipulation. These assumptions are more plausible for effort and spatial extent than for landings, while reported catch may be influenced by strategic considerations tied to subsidy eligibility.

Our work shows that total fishing effort decreases via two channels. First, total effort is reduced because many economic units cannot afford to operate without fuel subsidies. This is in line with observations from Aranceta-Garza and Cisneros-Montemayor [15], who calculated that fuel subsidies flipped operating profits from a net loss to a net gain in Mexico’s shrimp trawl fleet. The second channel is via the reductions in effort observed for economic units that do not exit the fishery. Together, these two channels result in net reductions in fishing effort and fisheries production, which should allow stocks to rebound.

These findings are consistent with theoretical predictions, which state that subsidizing variable inputs to fisheries increases effort and biomass extraction [1, 2]. However, our main contribution is that we provide empirical estimates that quantify the magnitude of the increases in fishing effort, fishing footprint, and fisheries production. These are important parameters that have direct relevance to the Agreement on Fisheries Subsidies at the World Trade Organization, which came into force on September 15, 2025 [6]. As countries begin

to plan the phasing-out of subsidy programs, they will require empirical estimates of the relationship between fuel subsidies and fishing effort or extraction. Our estimates may serve as starting points for the design and implementation of future subsidy reforms elsewhere.

Our retrospective analysis shows that around 27% of historical fishing effort in Mexico’s shrimp trawl fleet is attributable to subsidies. These findings imply that subsidy reform might generate large but localized environmental benefits. Limited availability of stock assessment data precludes precise statements about all relevant stocks, but we can put these numbers in context. For example, the biomass of the heavily fished blue shrimp (*Litopenaeus stylirostris*) stock in the Gulf of California is estimated at 30% below the biomass that would yield MSY (*i.e.*, $B/B_{MSY} = 0.7$; Cisneros-Mata [26]). It is therefore reasonable to expect that reducing fuel subsidies would support stock rebuilding, at least along the eastern Gulf of California.

Fuel subsidies are a direct driver of fishing pressure: they maintain marginal economic units in the fishery, intensify effort among those already active, and scale extraction with the size of the transfer. The effect is large—roughly a quarter of historical effort and two-fifths of landings in Mexico’s largest fleet—but unevenly distributed, resulting in concentrated reductions where subsidized effort is dense while leaving some of the most ecologically sensitive grounds untouched, because those waters are fished largely by economic units that never received support. Subsidy reform is therefore necessary to reduce total fishing effort, but it may not be sufficient. As the Agreement on Fisheries Subsidies moves from ratification to implementation [6], realizing its environmental promise will require pairing it with other fisheries management measures.

Declarations

Data and code - All data and code used in this manuscript is available on GitHub (https://github.com/jcvdav/mexican_subsidies).

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Funders had no say on the design and direction of the research.

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5 Methods

We are interested in studying how fuel subsidies affect fishing activity and production. Here, our unit of observation is an “economic unit”, a term used by the Mexican fisheries agency (CONAPESCA) to refer to an individual or firm that participates in a fishery. We combine administrative datasets on subsidy allocations by economic unit and vessel tracking data to construct a unique panel of annual subsidies and fishing activity by economic unit. The following subsections provide further details on data procurement, filters, and sample construction.

5.1 Datasets and their sources

We make use of six types of data to study the effects of fuel subsidies on fishing behavior. Historical subsidy allocations and a vessel registry allow us to identify subsidized economic units and their characteristics. Then, we use vessel tracking data and historical fisheries production data to derive our three main outcomes of interest: fishing time, fishing extent, and fisheries production. Finally, we also use historical diesel fuel prices and monthly indices for El Niño Southern Oscillation to include fuel costs and environmental variation as covariates. Each of these is described in detail below.

5.1.1 Subsidy allocations

Data on subsidy allocations to each economic unit come from CausaNatura, an NGO whose mission is to compile, procure, and make available administrative datasets relevant to environmental and natural resource management. We use the “*Padrón de beneficiarios de Combustibles*”, which was last updated on June, 2020. This administrative dataset contains information on the annual subsidy cap assigned to economic units fishing in Mexico during the 2011 - 2019 period ($n = 4,597$). From this information, we assign treatment status (subsidized or not subsidized) to all economic units in our sample (see subsection 5.2), and the

amount of fuel subsidy received by each.

5.1.2 Vessel registry

We use an official vessel registry with information for all large-scale fishing vessels that hold a fishing permit in Mexico, which was also provided by CausaNatura. The vessel registry includes unique vessel identifiers and economic unit identifiers (ownership), vessel dimensions (length overall, beam, draught, and gross tonnage), species-specific fishing permits granted, and engine characteristics (*e.g.* total engine power, type of fuel used by the engine, and engine model). The registry contains information for 3,093 vessels owned by 1,093 economic units. From these, 1,415 are licensed to use bottom trawl nets and 1,561 are licensed to participate in the shrimp fishery; 1,368 are licensed to use both (and are owned by 464 economic units).

5.1.3 Vessel tracking data

There are two general types of vessel tracking technologies: Automatic Identification Systems (AIS) and Vessel Monitoring Systems (VMS). AIS is designed as a vessel-to-vessel broadcast system intended to help avoid at-sea collisions between vessels [10]. VMS, on the other hand, is employed by governments to track vessels of interest, and a vessel’s position is broadcast directly to a central repository instead of to other vessels [27]. We use VMS data from Mexico’s satellite monitoring system of fishing vessels (*i.e.* SISMEP[28]). These data are publicly available and continuously updated at datos.gob.mx. The version we use was downloaded on June 15, 2024 [29]. These VMS data contain the timestamp, geographic location (latitude and longitude), and speed of 2,775 vessels between January 1, 2007 and Feb 29, 2024.

Regulations require all fishing vessels larger than 10.5 m in length overall and with an in-board motor of more than 80 horsepower to carry a Vessel Monitoring System (VMS)⁴.

⁴Regulatory text available at: <https://www.monitoreodeembarcaciones.com.mx/monitoreosatelital/QuienDebe.htm>

Failure to comply with this VMS requirement automatically disqualifies a vessel as eligible to receive any type of subsidy.

5.1.4 Fisheries production

Fisheries production data come from Mexico’s landing receipts, where fishers report their landings [30, 31]. As with the VMS requirement, failure to report catch makes a fisher ineligible to receive a subsidy. The dataset contains information on the identity of the economic unit and vessel landing the catch, the target species, and the amount (kg).

5.1.5 Fuel prices

We also compile price data for diesel fuel used by these economic units by combining two sources of information. The first one is reported by the Energy Information System (“Sistema de Información Energética”; “SIE”) and contains the national annual average price of diesel between 2011 - 2016, when fuels were subject to nation-wide price controls. Price controls were lifted in 2017, and fuel prices were determined by local supply and demand. The Energy Regulatory Commission (“Comisión Reguladora de Energía”; CRE) reports monthly state-level prices after 2017, which we use to calculate annual national averages for 2017 - 2019 period. We use Mexico’s consumer price index reported by the OECD to adjust prices to 2019 Mexican pesos.

5.1.6 Environmental covariates

The productivity of shrimp fisheries is known to be influenced by ENSO events [32]. We use the Mean NINO 3.4 index available from NOAA’s Physical Sciences Laboratory Climate Indices repository (Monthly Atmospheric and Ocean Time Series). We use monthly means to produce an annual mean value of ENSO 3.4, which we include as a time-varying covariate in some of our regressions.

5.2 Data processing

5.2.1 Sample construction

We limit our data to activity occurring between 2011 and 2019, the years for which subsidy allocation data are available. Additionally, we retain vessel tracks occurring at depths between 9.15 and 100 m deep (as indicated by GMED’s bathymetric dataset) because shrimp trawlers in Mexico are not allowed to fish shallower than 9.15 m deep and they operate at a maximum depth of 100 m [16]. Shrimp trawlers typically operate speeds between 1 and 5 knots⁵, so we also filter tracks based on their speed. These filters result in a total of 1,125 vessels belonging to 403 economic units. We further restrict the sample to economic units that are only licensed to fish for shrimp using trawl nets, leaving us with 341 economic units.

5.2.2 Outcomes of interest

Our first outcome of interest is fishing activity. We define it as time (hours) a vessel spent traveling at speeds between 1 and 5 knots in areas between 9.15 and 100 m depth. We calculate an economic unit’s total annual fishing hours as the sum of fishing hours across all their vessels.

Our second outcome of interest is the total extent of fishing grounds (km²) in which these economic units operate. We used a density-based spatial clustering algorithm (DBSCAN) to identify fishing grounds based on individual vessel positions. The algorithm was applied to all positions at the vessel-by-year level. The algorithm clusters points based on their distribution across space, given a minimum number of points per cluster and a maximum distance between points. We used a maximum distance of 50 km and a minimum of 6 points per cluster. Clusters thus represent the group of individual GPS coordinates that are associated with a fishing ground. Points without cluster membership were dropped. We then built a convex hull around each cluster and calculated its area. The total extent of fishing

⁵Catálogo de los Sistemas de Captura de las Principales Pesquerías Comerciales, available at: CONAPESCA

grounds of an economic unit was then calculated as the sum of all fishing grounds used by their vessels. For this portion of the analysis, geographic coordinates were reprojected onto a Mexico Lambert Conic Conformal projection (With EPSG code 6361).

Our third and last outcome of interest is the total amount of catch landed by each economic unit, which we derive from our fisheries production dataset. Our sample is therefore made up of large-scale economic units that target shrimp and carry VMS transponders. This group receives between 48.22% and 67.73% of the annual subsidies awarded to all industrial economic units fishing in Mexico. The final estimation sample is a panel of annual economic-unit fuel subsidy allocations (in 2019 MXP), time, extent, landings, and control variables such as fuel prices, total horsepower, number of active vessels owned by an economic unit, and environmental indices (*i.e.* NINO3.4 index). These data contain 2,942 observations attributed to 341 economic units between 2011 and 2019. Tables with summary statistics are included in the supplementary materials (Table S1).

5.3 Empirical strategy

5.3.1 Changes in subsidy status

Because roster turnover generates changes in subsidy status that are plausibly uncorrelated with unit-level determinants of the outcomes, we can use these quasi-random changes in subsidy status to test for changes in fishing behavior and fisheries production for economic units whose subsidy status changed at least once in our study period (2011-2019). We estimate the semi-elasticity (*i.e.* the % change in outcome of interest caused by change in subsidy status) of time fishing, fished area, and landings with respect to subsidy status in a two-way fixed-effects regression framework.

$$\mathbb{1}(y_{it} > 0) = \beta S_{it} + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \quad (1)$$

where $\mathbb{1}(y_{it} > 0) = 1$ if economic unit i has fishing hours, fished area, or landings reported for

year t . Then, S_{it} is a dummy variable indicating subsidy status, ω_i and $\lambda_{r(i)t}$ are fixed-effects by economic unit and by region-year. ($r(i)$ denotes the (time-invariant) region to which economic unit i belongs). Our parameter of interest is β , which captures the extensive margin effects. We also estimate a model where our outcome variables are simply y_{it} and $\log(y_{it})$, in which case β yields changes along the intensive margin and a semi-elasticity, respectively. Our supplementary materials include alternative specifications and sample definitions (See subsection A.3).

5.3.2 Changes in subsidy amount

Recall that our dataset has three types of economic units: those that were never subsidized, those that were subsidized at least one year, and those that were subsidized every year in our dataset. For the later two types, the *amount* of subsidy they receive varies by year (See Figure S2). This annual variation is due to budgetary constraints, which arise when CONAPESCA receives different amounts of funding in the annual federal budget or when funds are allocated to other programs (*e.g.* aquaculture development). We can use these quasi-random changes in subsidy amounts to test for changes in fishing behavior and fisheries production for economic units who were subsidized at least twice between 2011-2019. Like before, we estimate our coefficient of interest (this time an elasticity) in a two-way fixed effects framework with our estimating equation taking the form:

$$\log(y_{it}) = \beta \log(s_{it}) + \omega_i + \lambda_{r(i)t} + \epsilon_{it} \quad (2)$$

where $\log(y_{it})$ is the log-transformed outcome of interest and s_{it} is the subsidy amount allocated to economic unit i in year t , in 2019 Mexican pesos. ω_i and $\lambda_{r(i)t}$ are fixed-effects by economic unit and by region-year. The parameter of interest is β , which captures the elasticity of fishing with respect to subsidy amount.

5.3.3 An *impromptu* reform

Mexico eliminated fuel subsidies starting in 2020. We first test for changes in the probability of an economic unit exiting the fishery using the following specification:

$$E_{it} = \beta \text{Post}_t + \omega_i + \epsilon_{it} \quad (3)$$

where E_{it} is an indicator variable that takes a value of $E_{it} = 1$ if economic unit i exited the fishery at year t and is $E_{it} = 0$ otherwise. The variable Post_{it} takes a value of $\text{Post}_t = 1$ if the observation comes from the post-reform period (2020 and later) and $\text{Post}_{it} = 0$ otherwise. Finally, ω_i are fixed-effects by economic unit. The parameter of interest is β , which captures the average annual probability of an economic unit exiting the fishery in the post-reform period.

We then limit the sample to economic units that did not leave the fishery following the reform and estimate a model:

$$y_{it} = \beta \text{Post}_t + \omega_i + \epsilon_{it}$$

where y_{it} represents our three main outcomes of interest. This model measures the intensive margin effects of the subsidy reform.

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A Supplementary Materials

A.1 Subsidy program description

For the time period analyzed in this study (2011 - 2019), four related fuel subsidy programs in Mexican fisheries have been implemented: *PROCAMPO para vivir mejor* (2011 - 2012), *PROCAMPO Productivo* (2013), *Fomento a la productividad pesquera* (2014 - 2019) and *Subcomponente diesel marino* (2018). The operational rules of the fuel subsidy program in Mexican fisheries are as follows. The fuel subsidy program provides a 2-peso per liter subsidy over a portion of the total fuel used by a vessel, here termed the fuel cap of vessel i ($\hat{Q}_i$). As stated in the program's operational rules⁶, the subsidized portion of fuel for any diesel-consuming vessel is calculated using the following formula:

$$\hat{Q}_i = (MDL_i \times DPC_i) \times AF_i \quad (4)$$

Where $\hat{Q}_i$ represents the fuel cap on the subsidy program given to vessel i . MDL_i denotes the "Maximum Daily Liters" of vessel i , and is what the government expects the vessel's fuel consumption to be. DPC_i represents the "Days Per Cycle": the number of days a vessel is allowed to fish during a fishing season. The MDL_i is based on engine size (Figure S1), while DPC_i is determined by the fishery in which the vessel participates⁷. Finally, AF_i is an exogenous adjustment factor set by CONAPESCA and takes values between 0 and 1. This is independent of fishery, engine power, or stock status and is instead determined by budgetary constraints. The adjustment factor was typically set between 0.4 and 0.7, but local officials may downward adjust it. These variations in adjustment factor provide the source of variation that we will use to identify the effect of fuel fishery subsidies on exacerbating overfishing.

⁶See Section 4.1.2 of Acuerdo por el que se dan a conocer las Reglas de Operación de los Programas de la Secretaría de Agricultura, Ganadería, Desarrollo Rural, Pesca y Alimentación, available at https://dof.gob.mx/nota_detalle.php?codigo=5228713&fecha=30/12/2011#gsc.tab=0

⁷A fishery is defined as the combination of species and location. For example a vessel targeting tuna in the Pacific ocean is part of the Pacific tuna fishery.

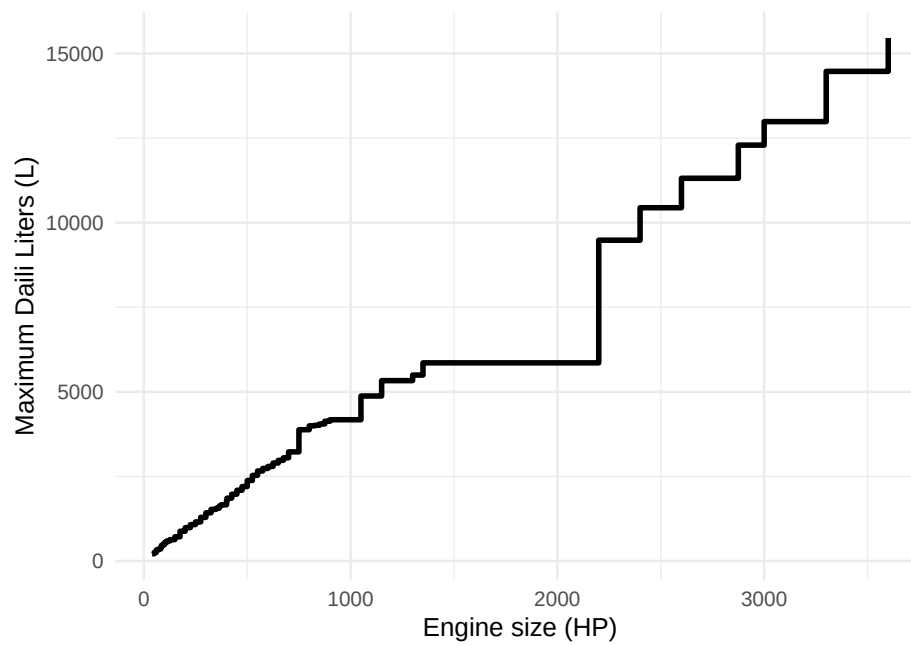


Figure S1: Expected daily fuel consumption for different engine powers. The x-axis shows engine power bins (in HP) as defined by CONAPESCA's operational rules. The y-axis shows the estimated maximum daily liters of fuel to be consumed for the corresponding engine power bin.

627 **A.2 Supplementary Figures and Tables**

Table S1: Summary statistics comparing the mean, standard deviation, and range of subsidy amounts and outcome variables across treatment statuses.

	treated	Mean	SD	Min	Max
Subsidy amount (2019 MXN)	Not subsidized	0.00	0.00	0.00	0.00
	Subsidized	790 634.44	1 511 080.57	1521.99	19 633 910.95
Fishing activity (hours)	Not subsidized	1177.34	2405.59	0.00	26 520.33
	Subsidized	4167.75	7053.72	0.00	65 835.78
Fished area (km ²)	Not subsidized	6168.02	9275.14	0.00	92 348.65
	Subsidized	17 966.41	16 374.48	0.00	116 771.44
Landings (Kg)	Not subsidized	9502.86	25 384.33	0.00	285 093.00
	Subsidized	67 204.62	101 070.81	0.00	1 099 556.00

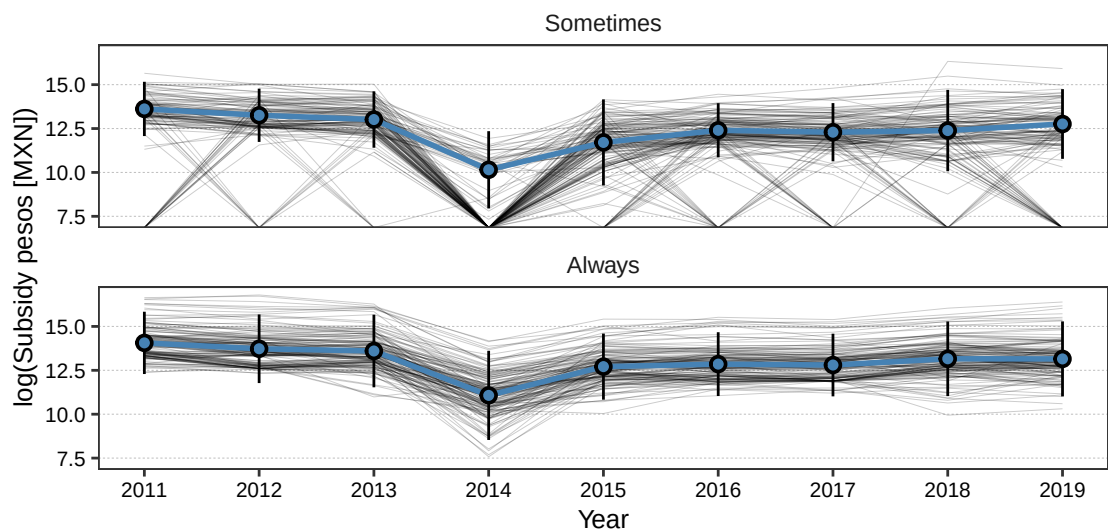


Figure S2: Change in the subsidy amount (Mexican Pesos) granted to each economic unit between 2011 and 2019. The top panel shows data for economic units that are subsidized at least once, the bottom panel shows data for economic units that are always subsidized in our period of study. Each thin black line corresponds to one economic unit. When a line touches the horizontal axis it implies it is not subsidized in that period. The overlaid points show mean $\pm$ sd. Note that not a single black line is perfectly horizontal, providing evidence of within-unit variations in subsidy allocations through time.

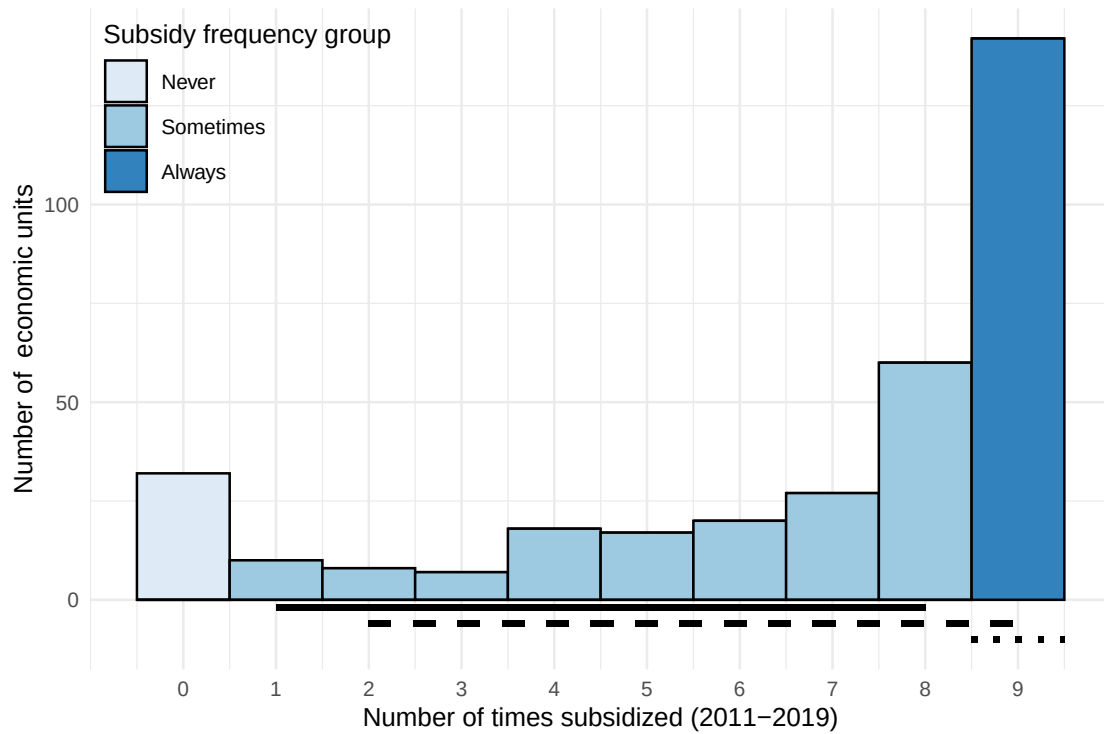


Figure S3: Histogram of frequency with which economic units are subsidized (2011-2019). A value of $N = 0$ along the horizontal axis implies never subsidized, while $N = 9$ implies always subsidized. Our main semi-elasticity estimates use economic units sometimes subsidized (i.e. $N = 1-8$). Our main elasticity estimates use all economic units subsidized $N \geq 2$ times.

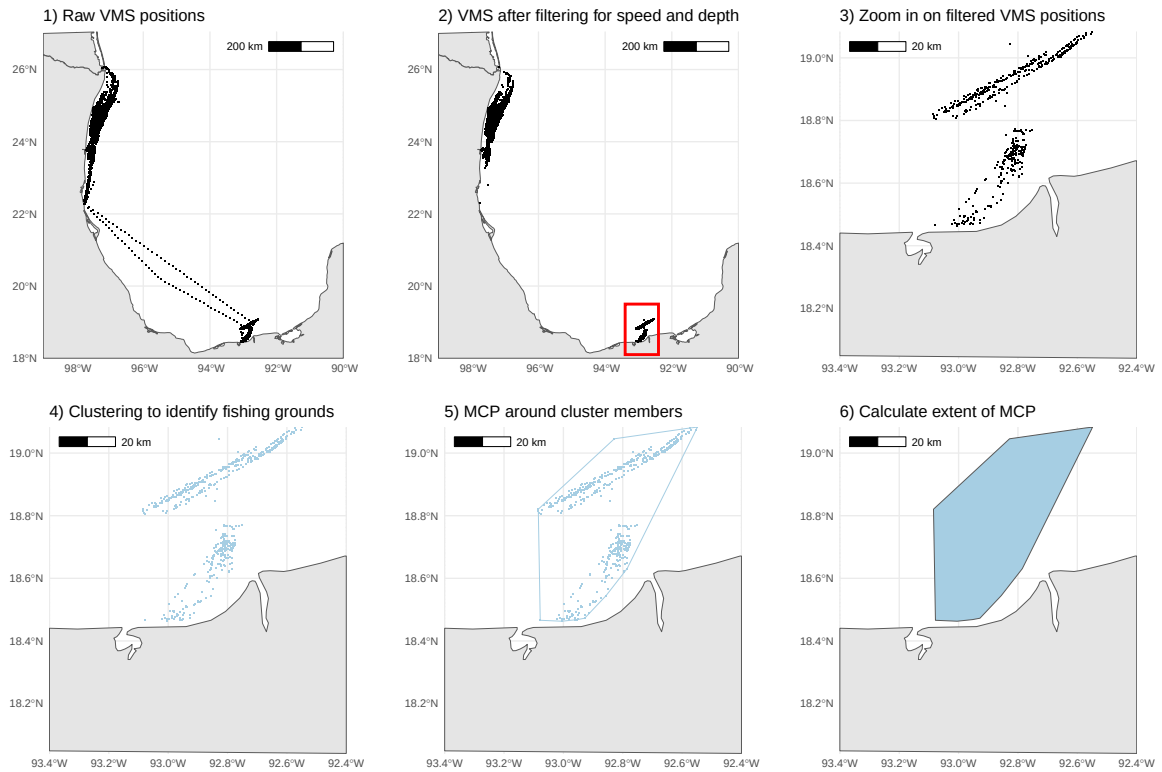


Figure S4: Example showing the process of identifying fishing grounds. Panel 1) shows raw VMS positions for a single vessel. Panel 2) shows VMS positions after filtering for speed and depth parameters. Panel 3) zooms in on the southernmost fishing grounds highlighted in the red square of panel 2). Panel 4) shows the output from clustering. Panel 5) shows the minimum convex polygon (MCP) around cluster members and Panel 6) shows the extent of each fishing ground.

A.3 Robustness tests

A.3.1 Responses to changes in subsidy status

We inspect the robustness of our main-text estimates by performing a series of tests using different specifications and samples. Our first test drops all fixed-effects, and instead incorporates covariates for number of vessels in 2011, total engine power in 2011, log-price of diesel fuel, and nino3.4 index interacted by region. This specification produces similar coefficient estimates. The second test uses the same specification as our main-text estimates, but instead modifies the sample. Recall that our main-text sample excludes economic units never ($N = 32$) and always ($N = 142$) subsidized between 2011 and 2019. This second test includes all economic units, and yields similar coefficient estimates. Our final test removes economic units that were listed in the “fleet modernization” roster, meaning that, in addition to fuel subsidies, they received support to purchase new engines. Removing these economic units yields similar coefficient estimates, leaving our conclusions unchanged. We perform the same battery of tests for our extensive margin estimates (Figure S5 and Table S2), intensive margin estimates (Figure S6 and Table S3), and semi-elasticity estimates (Figure S7 and Table S4).

Finally, because economic units can come in and out of the roster multiple times, we also estimate a difference-in-differences estimator robust to treatment switching based on de Chaisemartin and D’Haultfoeuille [20]. We run the same regression specification as in the main text and simultaneously check for lack of pre trends by estimating an event-study with two placebo periods and three effect periods. Extending the number of placebos to more than two periods would have yielded estimates with very few switchers. We use the DIDmultiplegtDYN package in R [33, 34]. Because the treatment is non-absorbing, we should expect to only see an effect for period = 1, the period where an economic unit receives a subsidy. This is precisely what we observe, and the magnitude and direction of coefficients remains similar. The results are presented in Figure S8 and Figure S9.

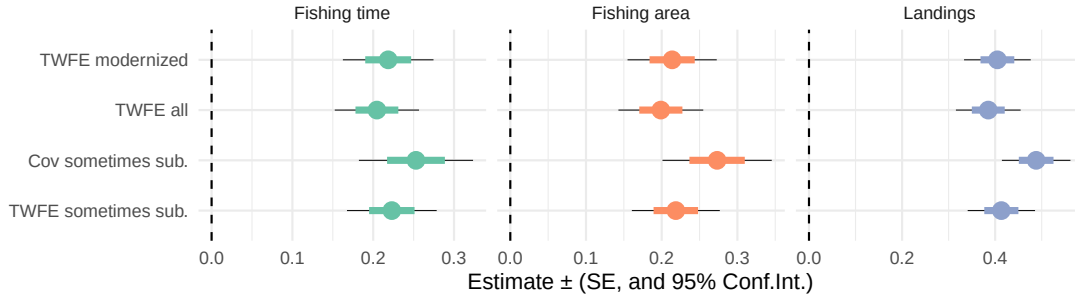


Figure S5: Coefficient estimates for the extensive margin on time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes economic units that were recipients of the fleet modernization subsidies. See Table S2 for more details.

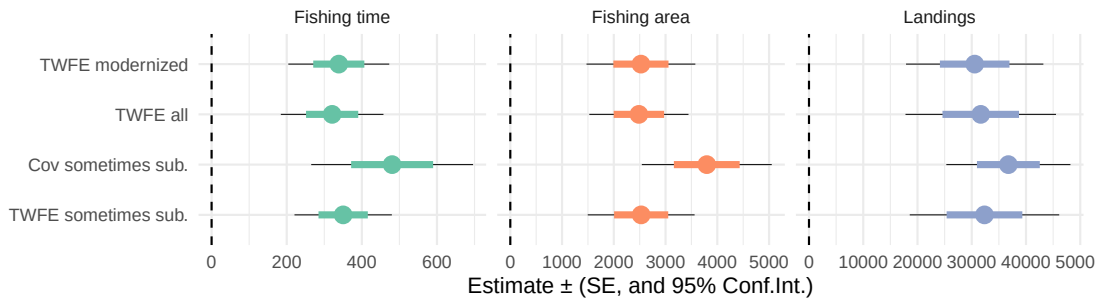


Figure S6: Coefficient estimates for the intensive margin (in levels) of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes economic units that were recipients of the fleet modernization subsidies. See Table S3 for more details.

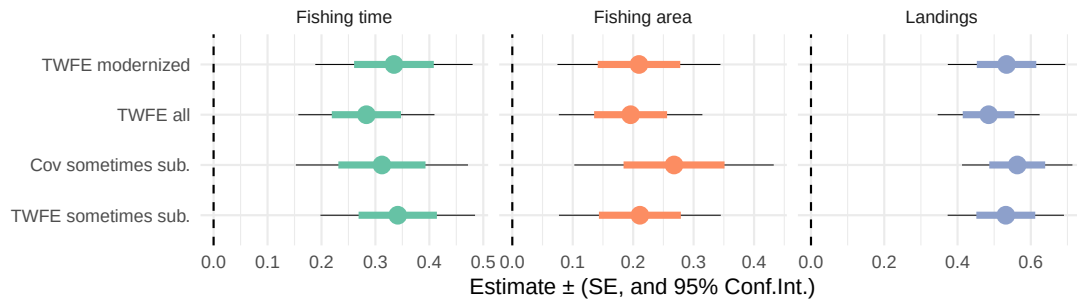


Figure S7: Coefficient estimates for the semi-elasticities of time fishing, fishing area, and landings with respect to subsidy status for different specifications and samples. Points are coefficient estimates, colored lines show panel-robust standard errors, and black lines show 95% confidence intervals. “TWFE sometimes sub.” refers to the main-text estimates, “Cov sometimes sub” refers to estimates for a model specification that swaps fixed-effects for covariates. “TWFE all” refers to the same two-way fixed effects specification as in the main text, but this time including all economic units regardless of their subsidy frequency and status. “TWFE modernized” removes economic units that were recipients of the fleet modernization subsidies. See Table S4 for more details.

Table S2: Effect of receiving a fuel subsidy on time fishing (hours) >0, fishing area (km²) >0, and landings (kg) >0.

	Fishing time	Fishing area	Landings
A) Main text specification			
Received subsidy	0.223 (0.028)***	0.219 (0.029)***	0.413 (0.037)***
N	1431	1431	1431
R^2 Adj	0.531	0.547	0.597
B) Covariates but no fixed effects			
Received subsidy	0.253 (0.036)***	0.273 (0.037)***	0.488 (0.037)***
N	1431	1431	1431
R^2 Adj	0.206	0.188	0.373
C) Main text specification with all units			
Received subsidy	0.205 (0.027)***	0.199 (0.028)***	0.385 (0.035)***
N	2941	2941	2941
R^2 Adj	0.520	0.588	0.703
D) Main text specification without modernized units			
Received subsidy	0.219 (0.028)***	0.214 (0.030)***	0.405 (0.036)***
N	1411	1411	1411
R^2 Adj	0.531	0.547	0.603

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1A. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S3: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km²), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Received subsidy	350.269 (65.646)***	2528.340 (522.980)***	32347.423 (6977.661)***
N	1431	1431	1431
R^2 Adj	0.924	0.792	0.710
B) Covariates but no fixed effects			
Received subsidy	480.619 (109.122)***	3794.673 (635.756)***	36753.146 (5792.281)***
N	1431	1431	1431
R^2 Adj	0.752	0.488	0.377
C) Main text specification with all units			
Received subsidy	320.930 (69.474)***	2484.836 (487.043)***	31658.543 (7053.213)***
N	2941	2941	2941
R^2 Adj	0.956	0.863	0.915
D) Main text specification without modernized units			
Received subsidy	338.652 (68.042)***	2523.971 (532.576)***	30543.733 (6410.022)***
N	1411	1411	1411
R^2 Adj	0.922	0.793	0.712

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1B. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

Table S4: Effect of receiving a fuel subsidy on time fishing (hours), fishing area (km²), and landings (kg).

	Fishing time	Fishing area	Landings
A) Main text specification			
Received subsidy	0.342 (0.073)***	0.211 (0.068)***	0.532 (0.080)***
N	1290	1235	1192
R^2 Adj	0.726	0.699	0.757
B) Covariates but no fixed effects			
Received subsidy	0.312 (0.081)***	0.267 (0.084)***	0.563 (0.076)***
N	1292	1240	1196
R^2 Adj	0.305	0.245	0.294
C) Main text specification with all units			
Received subsidy	0.283 (0.064)***	0.196 (0.060)***	0.485 (0.071)***
N	2723	2629	2530
R^2 Adj	0.791	0.783	0.846
D) Main text specification without modernized units			
Received subsidy	0.334 (0.074)***	0.210 (0.068)***	0.534 (0.081)***
N	1272	1217	1173
R^2 Adj	0.727	0.701	0.756

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 1C. Panel B) uses the same sample of vessels subsidized at least once, but removes all fixed effects and adds covariates for number of vessels, total engine power, log-price of diesel fuel, and nino3.4 index interacted by region. Panel C) uses the same two-way fixed effects estimation as in A), but includes all vessels in our sample, regardless of number of times subsidized. Panel D) uses the same two-way fixed effects estimation as in A), but removes vessels that received a fleet modernization subsidy.

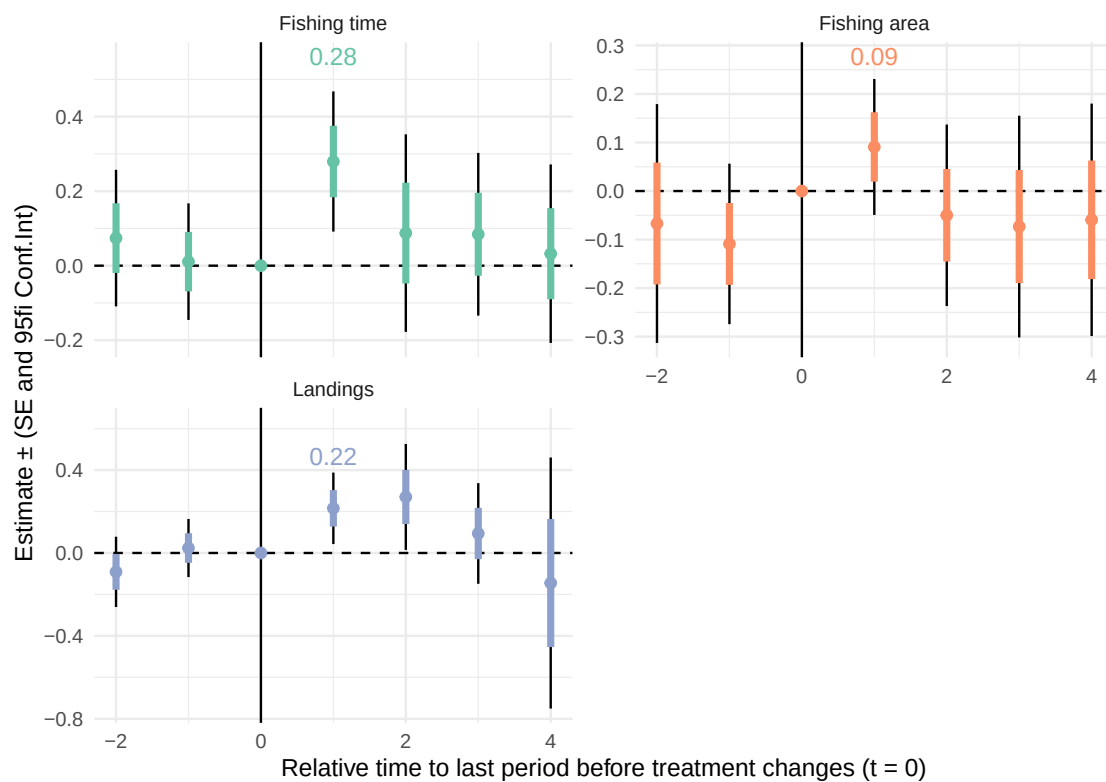


Figure S8: Event study plot using a difference-in-differences estimator that is robust to treatment switching. Outcomes are log-transformed in all regressions.

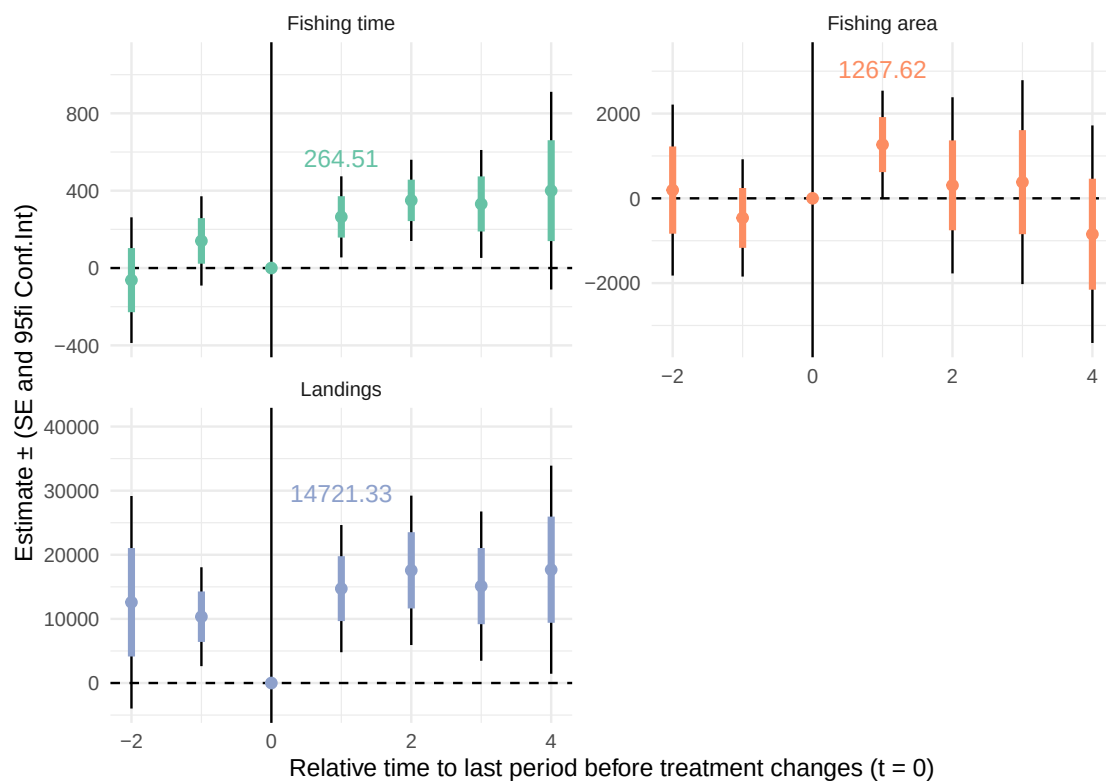


Figure S9: Event study plot using a difference-in-differences estimator that is robust to treatment switching. Outcomes are in levels in all regressions.

A.3.2 Responses to changes in subsidy amounts

We inspect the robustness of our main-text elasticity estimates by performing a series of tests using different specifications and samples. The data in Table S5 provides model fit statistics and sample sizes, but coefficients are more easily visualized in Figure S10. Our main-text sample includes all economic units subsidized at least twice between 2011 and 2019 ($N = 297$). Our first test modifies the sample to include only economic units that were always subsidized during the same period ($N = 142$). This sub-sample produces similar coefficient estimates, shown in Panel B of Table S5. The next test does the opposite, and excludes economic units that were always subsidized. The results are shown in Panel C of Table S5, and show coefficient estimates similar to those reported in our main text. As before, the third test removes economic units that participated in the fleet modernization program. This also produces similar coefficient estimates for all outcome variables, as shown in Panel D of Table S5. Finally, we drop our fixed-effects by year and economic unit and instead include the same battery of covariates as before. This specification uses the same sample as our main-text estimates, and results in higher estimates of elasticity of fishing time, fishing area, and landings at 0.2, 0.19, and 0.37, respectively. These estimates are 40-90% larger than our main-text estimates, and are shown in Panel D of Table S5.

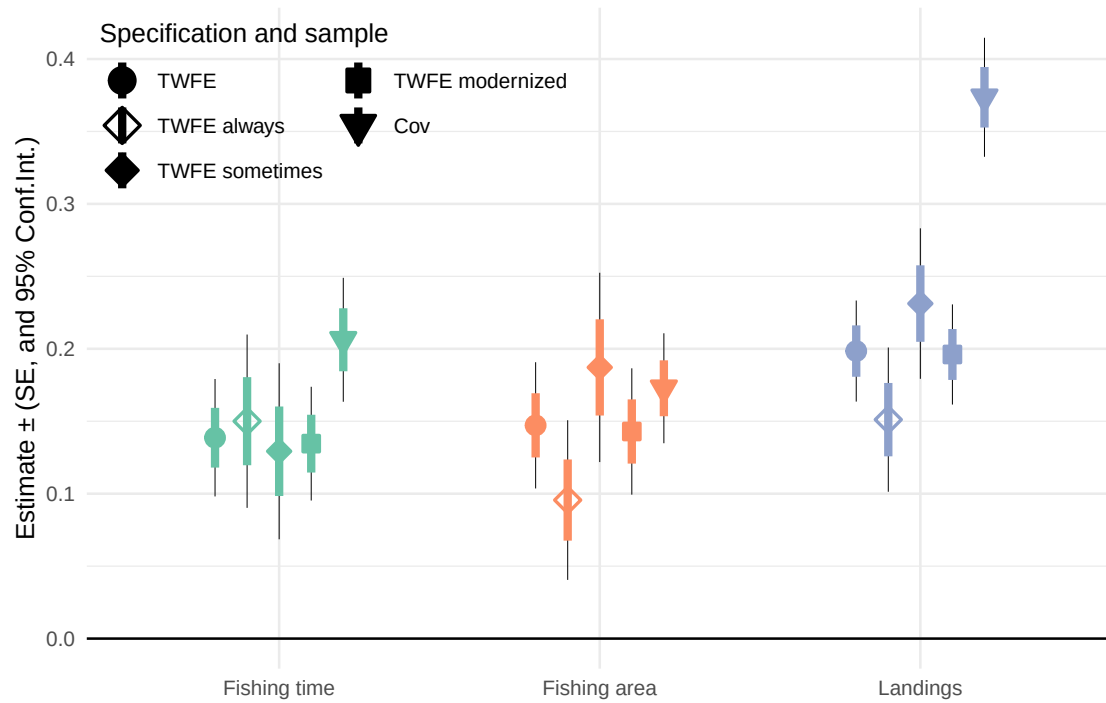


Figure S10: Coefficient estimates for the elasticities of time fishing, fished area, and landings with respect to subsidy amount. Points are coefficient estimates, colored lines show standard errors, and black lines show 95% confidence intervals. The main sample combines all economic units subsidized at least twice between 2011 and 2019. Alternative samples, labeled “Modernized” “Sometimes” and “Always”, restrict the sample to economic units that were not part of fleet modernization subsidies, or that are sometimes and always subsidized in the same period, respectively. One-way fixed-effect specifications (labeled “OWFE”) drop year-by-region fixed effects and use annual log-transformed mean national fuel prices, the NINO3.4 index values, and a dummy variable for 2014.

Table S5: Elasticity estimates for time fishing (hours), fishing area (km²), and landings (kg) with respect to changes in subsidy amount.

	Fishing time	Fishing area	Landings
A) Main text specification			
log(subsidy amount[MXP])	0.139 (0.021)***	0.147 (0.022)***	0.198 (0.018)***
N	2240	2202	2246
R^2 Adj	0.850	0.818	0.876
B) Always subsidized			
log(subsidy amount[MXP])	0.150 (0.030)***	0.096 (0.028)***	0.151 (0.025)***
N	1278	1271	1278
R^2 Adj	0.884	0.828	0.905
C) Sometimes subsidized			
log(subsidy amount[MXP])	0.129 (0.031)***	0.187 (0.033)***	0.231 (0.026)***
N	962	931	968
R^2 Adj	0.749	0.746	0.809
D) Removing modernized			
log(subsidy amount[MXP])	0.135 (0.020)***	0.143 (0.022)***	0.196 (0.018)***
N	2208	2170	2214
R^2 Adj	0.850	0.819	0.875
D) Covariates but no fixed effects			
log(subsidy amount[MXP])	0.206 (0.022)***	0.173 (0.019)***	0.374 (0.021)***
N	2242	2206	2247
R^2 Adj	0.517	0.374	0.540

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is an economic unit by year. Numbers in parentheses are cluster-robust standard errors, clustered at the economic-unit level. Panel A) shows the same information as in Table 2. Panel B) restricts the sample to economic units always subsidized. Panel C) restricts the sample to economic units sometimes subsidized. Panel D) removes economic units that received fleet modernization subsidies. Panel E) uses the same sample of vessels, but removes all fixed effects and adds covariates for number of vessels, total engine power, and nino3.4 index interacted by region.